Predicting Antibiotic Resistance in Bacteria Using Machine Learning

A Genomic Data Science Case Study - Klebsiella pneumoniae Meropenem Resistance

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Abstract

This project builds a complete machine learning pipeline to predict antibiotic resistance in *Klebsiella pneumoniae* bacteria using their DNA sequences. We downloaded 726 bacterial genomes from the BV-BRC global genomic database, extracted DNA pattern features (k-mers), and trained four machine learning models to classify each genome as Resistant or Susceptible to meropenem - a last-resort antibiotic. Our best model, Linear SVM, achieved an AUC of 96.24% and Recall of 88.65% under rigorous cross-validation. However, a critical diagnostic investigation revealed that 98% of genome pairs shared near-identical DNA profiles, indicating the dataset is dominated by clonal outbreak strains rather than genuinely diverse isolates. This finding - phylogenetic data leakage explains the inflated metrics and represents the most important methodological contribution of this work. The pipeline, diagnostic framework, and lessons learned are now being applied to a Nigeria-specific resistance prediction study where dataset diversity has been validated.

1. The Problem We Are Solving

Klebsiella pneumoniae is a bacterium responsible for some of the most dangerous hospital infections in the world including pneumonia, bloodstream infections, urinary tract infections, and surgical wound infections. It is particularly dangerous in hospitals across Nigeria and other African countries where infection control resources are limited.

Meropenem is a carbapenem antibiotic - a last-resort drug used when almost everything else has failed. When *K. pneumoniae* becomes resistant to meropenem (called carbapenem-resistant K. *pneumoniae* or CRKP), clinicians have almost no remaining treatment options. The patient's survival depends on identifying this resistance quickly and switching to alternative strategies.

The traditional way to detect antibiotic resistance is to grow the bacteria in a laboratory with the antibiotic and wait to see whether it survives. This process takes 48 to 72 hours in ideal conditions and in many Nigerian hospitals, significantly longer. In a critically ill patient, this delay can be fatal.

Our approach: read the DNA sequence of the bacterium and use machine learning to predict its resistance profile in minutes, without waiting for a laboratory culture result.

2. Dataset and Data Acquisition

We sourced our data from the BV-BRC (Bacterial and Viral Bioinformatics Resource Center), formerly known as PATRIC - the world's largest publicly available bacterial genomics database. BV-BRC contains whole genome sequences for thousands of bacterial isolates, each linked to laboratory-confirmed antibiotic susceptibility results.

Data acquisition steps:

- Queried BV-BRC API for all *K. pneumoniae* genomes with meropenem susceptibility data
- Retrieved 5,000 records; 2,184 had confirmed resistance labels (Resistant or Susceptible)
- Removed 2 Intermediate-labeled records, clinically ambiguous for binary classification
- Downloaded whole genome FASTA sequences for 743 labeled genomes using batch API calls
- Matched downloaded sequences with resistance metadata, 726 genomes successfully matched
- Removed duplicate and contradictory-labeled genomes - final clean dataset: 726 genomes

Final dataset composition:

Class	Count	Percentage	Clinical Meaning
Susceptible	514	70.8%	Meropenem would work for this patient
Resistant	212	29.2%	Meropenem would fail - needs alternative treatment
Total	**726**	**100%**	

The 70/30 class imbalance is clinically realistic - not every *K. pneumoniae* infection involves a resistant strain. However this imbalance requires careful handling during model training to prevent the model from defaulting to always predicting Susceptible.

3. Feature Engineering - Turning DNA into Numbers

Machine learning models cannot read DNA sequences directly, they need numbers. Our feature engineering process converts raw DNA text into a numerical matrix that models can learn from.

What are k-mers?

A k-mer is a short DNA substring of fixed length k. By sliding a window of length k across a genome and counting how often each unique substring appears, we create a frequency profile for that genome. These frequency profiles become the features for our machine learning models.

Example: for the sequence ATGCTT with k=3:

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ATG → TGC → GCT → CTT   (4 trinucleotides from a 6-base sequence)
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We used k=6 (hexanucleotides), producing 18,016 unique 6-mer features per genome. The resulting feature matrix has 726 rows (one per genome) and 18,016 columns (one per unique 6-mer). Because most k-mers appear rarely in any given genome, this matrix is stored in sparse format, a memory-efficient representation that only records non-zero values.

Why k-mers work biologically:

Antibiotic resistance genes have characteristic DNA sequences that produce distinctive k-mer patterns. For example, the KPC carbapenemase gene responsible for meropenem resistance in *K. pneumoniae* contains specific hexanucleotide sequences not commonly found in susceptible strains. Without explicitly telling the model which genes to look for, k-mer frequency profiles allow it to discover these patterns from the data.

4. Critical Finding- Phylogenetic Data Leakage

Before building our final models, we made an important and unexpected discovery that fundamentally changed how we interpret our results. This investigation is the most scientifically valuable contribution of this project.

The initial red flag:

A naive train-test split produced 100% classification accuracy - an impossibly perfect result that immediately suggested something was wrong. Even after implementing 5-fold stratified cross-validation, the AUC remained at 94.41%, suspiciously high for a complex biological problem.

The investigation:

We computed pairwise cosine similarity between the k-mer profiles of 100 randomly sampled genomes. Cosine similarity measures how alike two genomes are based on their DNA pattern frequencies, ranging from 0 (completely different) to 1.0 (identical).

Finding	Value	What It Means
Highly similar pairs (>0.99 similarity)	4,851 of 4,950	98% of all pairs checked
Identical genome pairs (1.0 similarity)	5+ pairs	Exact duplicates present in dataset

Distinct clusters at 99% similarity	17 clusters	From 200 genomes sampled
Distinct clusters at 95% similarity	2 clusters	200 genomes = only 2 biological groups

The biological explanation:

K. pneumoniae resistant to meropenem spreads through hospital outbreaks. A single resistant strain infects many patients in the same ward over weeks or months. Each patient's isolate gets cultured, sequenced, and submitted to BV-BRC independently but they are all essentially the same bacterium with tiny differences. Our 726 genomes likely represent only 17 to 50 truly distinct bacterial strains. The other 700+ are near-identical copies from the same outbreak clusters.

What this means for our results:

Our models are not learning generalizable antibiotic resistance mechanisms. They are memorizing strain-specific DNA fingerprints. When a strain appears in both the training and test sets (as near-identical genomes do), the model achieves high accuracy by recognition rather than biological understanding. This is called phylogenetic data leakage and it is one of the most common but least discussed problems in biological machine learning. Most papers never check for it. We caught it.

5. Model Training and Evaluation

Despite the clonal structure, we completed the full multi-model pipeline to demonstrate methodological rigour and provide a benchmark for future work on diverse datasets. All models were evaluated using 5-fold stratified cross-validation with SMOTE applied strictly within training folds.

Why SMOTE inside cross-validation folds?

SMOTE (Synthetic Minority Oversampling Technique) creates artificial Resistant genome profiles to balance the 70/30 class imbalance. Critically, we apply SMOTE only to training data within each fold never to test data. Applying SMOTE before splitting would allow synthetic samples to leak information between train and test sets, further inflating metrics.

Results - k=6 features with SMOTE:

Model	AUC-ROC	Recall	Precision	Specificity	F1	MCC
Logistic Regression	94.16%	82.98%	78.13%	90.26%	80.32%	0.72
Linear SVM ★	**96.24%**	**88.65%**	**90.25%**	**95.91%**	**89.36%**	**0.85**
Random Forest	94.63%	79.71%	79.13%	91.06%	79.07%	0.71
XGBoost	96.11%	89.08%	80.11%	90.66%	84.13%	0.78

★ Best performing model across all metrics

6. Clinical Interpretation

For every 100 truly resistant *K. pneumoniae* infections, our Linear SVM model correctly identifies 89 of them (88.65% recall). The remaining 11 patients would receive meropenem, a drug that will not work for them potentially leading to clinical deterioration before the error is caught.

For every 100 truly susceptible infections, the model correctly clears 96 of them (95.91% specificity). Only 4 patients would be unnecessarily escalated to alternative antibiotics, which are often more expensive, more toxic, or harder to access.

Important caveat: These performance figures are inflated by the clonal dataset structure described in Section 4. Real-world performance on genuinely novel strains would likely be lower. Clinical deployment would require validation on a diverse, phylogenetically independent dataset.

7. Honest Limitations

- Phylogenetic clonal bias: 98% of genome pairs share >0.99 DNA similarity. Models are memorizing strain fingerprints rather than learning generalizable resistance mechanisms. Standard cross-validation does not resolve this - phylogeny-aware splitting would be required for unbiased evaluation.
- Global dataset, not Nigerian-specific: All genomes sourced from BV-BRC, dominated by European and Asian isolates. Performance on Nigerian clinical *K. pneumoniae* strains is unknown.
- Single antibiotic focus: This pipeline addresses meropenem resistance only. Clinical utility requires multi-drug resistance profiles, which would require separate models per antibiotic.
- No SHAP interpretation completed: Feature importance via SHAP values was not completed in this version and is planned as future work.

8. Future Work

- k=8 feature comparison: Evaluate whether expanding from 6-mer to 8-mer features improves resistance classification under clonal conditions - and whether longer k-mers capture resistance gene signatures not detectable at k=6.
- Phylogeny-aware train-test splitting: Implement genome clustering before splitting to ensure clonally related strains never appear in both training and test sets.
- SHAP biological interpretation: Apply SHAP values to identify the most predictive k-mers and map them to known resistance genes using BLAST.
- African isolate validation: Filter dataset for African-origin isolates and evaluate whether a model trained on African strains outperforms a globally trained model on African clinical samples.
- Future Horizons: Adapting this genomic machine learning framework to map multi-drug resistance profiles in Nigerian clinical isolates.

9. Technical Stack

Component	Tool / Library	Purpose
Data acquisition	BV-BRC REST API, requests	Download AMR metadata and genome sequences
Sequence handling	Biopython (SeqIO)	Parse FASTA files, genome reconstruction
Feature extraction	scikit-learn CountVectorizer	k-mer extraction and sparse matrix creation
Sparse storage	scipy.sparse	Memory-efficient feature matrix storage
Class imbalance	imbalanced-learn SMOTE	Synthetic minority oversampling in training folds
Models	scikit-learn, XGBoost	LR, Linear SVM, Random Forest, XGBoost
Evaluation	scikit-learn metrics	AUC, Recall, Precision, Specificity, F1, MCC
Similarity analysis	cosine_similarity (sklearn)	Phylogenetic leakage investigation
Clustering	AgglomerativeClustering	Strain diversity quantification
Visualization	matplotlib, seaborn	Plots, heatmaps, ROC curves, confusion matrices

10. Key Takeaways

- A complete end-to-end genomic ML pipeline was built from raw API data to trained models demonstrating the full bioinformatics workflow from data acquisition through feature engineering to clinical evaluation.

- High accuracy is not always what it seems. The most important lesson from this project is methodological: always investigate why your model performs well, not just how well it performs.
- Phylogenetic data leakage is a real and serious problem in microbial genomics ML that standard cross-validation does not catch. This finding adds genuine value to the literature.
- Linear SVM is the optimal architecture for high-dimensional sparse k-mer data, outperforming Random Forest and XGBoost across every metric on this dataset.
- SMOTE applied within cross-validation folds successfully improved recall from 70.28% (baseline Random Forest without SMOTE) to 88.65% (Linear SVM with SMOTE) a clinically meaningful gain.

All code available at: github.com/Imoleoluwanimi | Contact: abioyedelight2017@gmail.com